

Question	Answer	Mark	Mark scheme	Additional guidance
16	$10\sqrt{2}$	P1	for process to find total surface area of sphere, eg $75\pi \div 3 \times 8 (= 200\pi)$ or $75 \div 3 \times 8 (= 200)$ or for setting up an equation, eg $\frac{3}{8} \times 4\pi r^2 = 75\pi$ or $\frac{3}{8} \times 4r^2 = 75$	Could work without π for the three P marks
		P1	for process to find r^2 , eg $(r^2 =) \frac{200\pi}{4\pi}$ oe or $(r^2 =) \frac{75\pi}{4\pi} \times \frac{8}{3}$ oe or $r^2 = "50"$	
		P1	for process to find radius, eg $(r =) \sqrt{\frac{200\pi}{4\pi}}$ oe or $\sqrt{50}$ or $5\sqrt{2}$ or for $(2r =) = \sqrt{200}$	
		A1	cao	